Where Muon's Advantage Lives

Small-scale causal localization of an optimizer's effect on learned features

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INTERACTIVE PAGE

IN SHORT

The features Muon and AdamW learn agree as well as two AdamW seeds agree with each other. Muon's training gain, and its more reproducible features, come from one place: the attention **value circuit** $W_V W_O$.

§1 The question

Muon replaces each momentum update $M = U\Sigma V^T$ with its nearest orthogonal matrix, $W \leftarrow W - \eta \cdot UV^T$. It trains language models faster than AdamW and leaves a fingerprint on the weights: flatter spectra, higher stable rank.

Q1 Does the optimizer change **which features** a model learns?
Q2 **Where** does Muon's gain live, and is the rank fingerprint the cause?

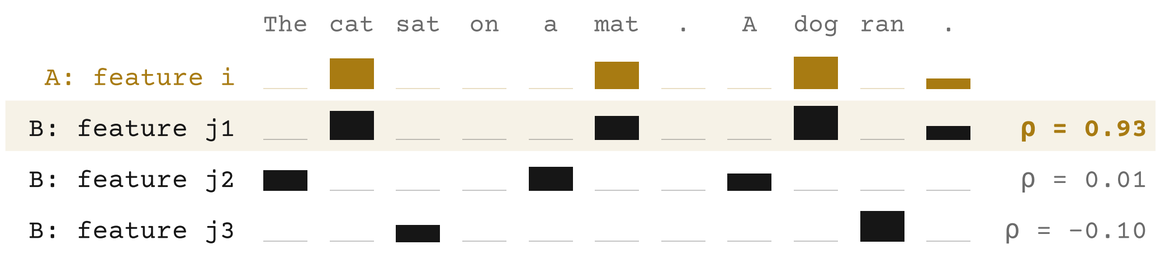
§2 Setup

124M GPT-2-class model (12 layers, d = 768, 6 heads, QK-norm) on FineWeb, 2.67B tokens, same data order. Each optimizer tuned; three seeds per arm.

`final val loss` **Muon 3.292** AdamW 3.350

§3 Comparing features

The same top-k sparse autoencoder on layer 6 of every model (12,288 features, k = 32). Decoder directions don't align even across seeds (mean best cosine 0.14), so we match **behaviour**: features match if they fire on the same tokens.



Toy example. Score = best Pearson ρ with any feature of the other model, over 1M shared held-out tokens.

— **Feature match:** that score, averaged over live features, symmetrized
— **Reproducibility:** feature match between two seeds of one arm
— At matched loss (Muon step 4,375, AdamW 5,100); floor 0.024

§4 Swapping circuits

An attention head factors into a **scoring** circuit $W_Q W_K^T$ (where to look) and a **value** circuit $W_V W_O$ (what gets written). Architecture and data fixed; Muon on some matrices, AdamW on the rest.

arm	Muon on	QK rank	OV rank
full AdamW	none	6.7	31.5
muon-QK	W_Q, W_K	**21.0**	28.7
muon-VO	W_V, W_O	5.5	**76.2**
muon-attn	W_Q, W_K, W_V, W_O	21.3	76.6
full Muon	all hidden	n/m	64.7

Stable rank of each circuit after training (whole layer, mean over layers); n/m = not measured. Fig. 3 adds muon-MLP and the complement (Muon on all but W_V, W_O).

§5 The features match at the seed ceiling

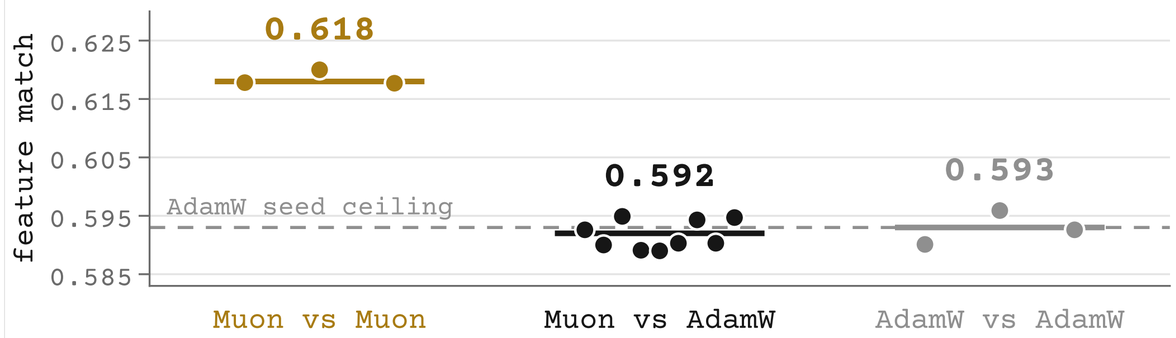


Fig. 1 Layer-6 feature match at matched loss, one dot per seed pair. Crossing the optimizer (0.592) costs no more than crossing the seed (0.593). Muon's own seeds agree more (0.618).

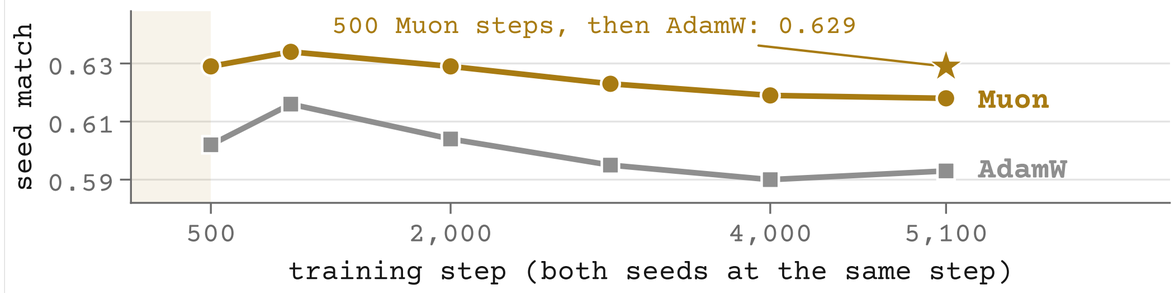


Fig. 2 The reproducibility gap is set by step 500; 500 Muon steps followed by AdamW keep it (0.629).

Geometry does change: Muon's stable ranks are 1.6–5.3× AdamW's and its residual stream is 5–7× larger.

§6 The value circuit is the lever

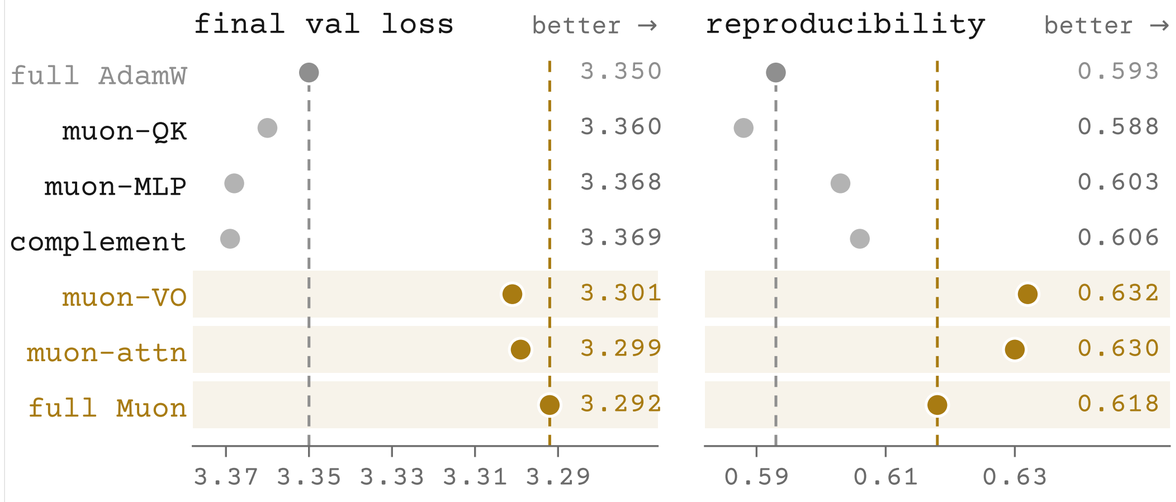


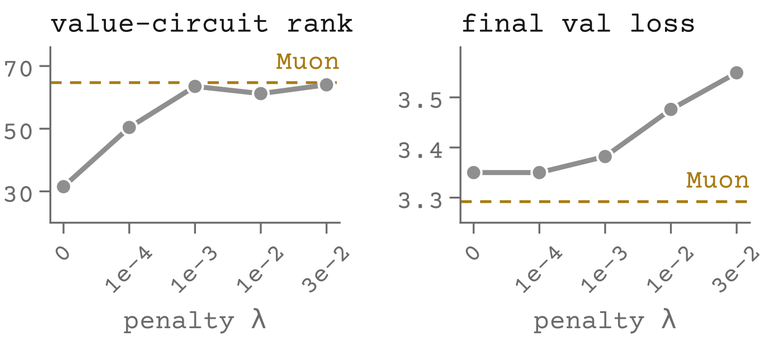
Fig. 3 Every arm, three seeds. Shaded rows put Muon on $W_V W_O$. Dashed lines: full AdamW and full Muon.

— **Sufficient.** muon-VO recovers 86% of the loss gap† and is the most reproducible arm (0.632).
— **Necessary.** Muon everywhere *except* $W_V W_O$ loses the whole gain (3.369, behind AdamW).
— **Not the MLP.** muon-MLP also ends behind AdamW (3.368).
— **A passenger.** muon-QK triples scoring rank and buys nothing.

† From unrounded three-seed means; the rounded losses shown give 84.5%.

§7 Three controls

RANK IS NOT THE KNOB



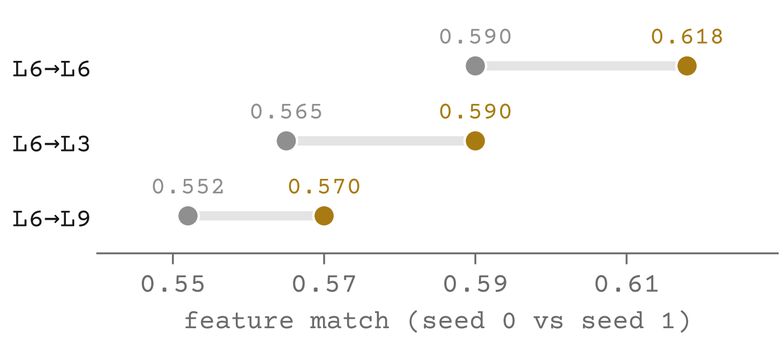
AdamW with a penalty on σ_1 of the value circuit reaches Muon's rank (64.7), but its loss only gets worse.

HOLDS ACROSS HEAD COUNTS

heads	Muon	AdamW	cross
3	0.625	0.594	0.595
6	0.618	0.593	0.592
12	0.611	0.573	0.577

Feature match at each head count, width fixed. Cross-optimizer stays at the AdamW ceiling; Muon stays more reproducible.

THE METRIC CAN MOVE



One seed pair throughout. Changing layer drops the score 0.03–0.05; crossing the optimizer drops it 0.001.

Conclusions

01 Crossing the optimizer costs no more feature agreement than crossing the seed. Muon's features are **more seed-reproducible**, from step 500.

02 Muon's advantage lives in the **value circuit** $W_V W_O$: sufficient, and necessary for the loss gain.

03 The rank fingerprint is a **passenger**, not the cause.

Scope: one 124M bench with 350M checks, one SAE family (top-k, firing-pattern matching), one dataset.

maincodemax.github.io/muon-feature-study

Jordan et al. 2024, Muon · Bernstein & Newhouse 2024, Old optimizer, new norm · Gao et al. 2024, Scaling and evaluating sparse autoencoders · Elhage et al. 2021, A mathematical framework for transformer circuits · Penedo et al. 2024, FineWeb